

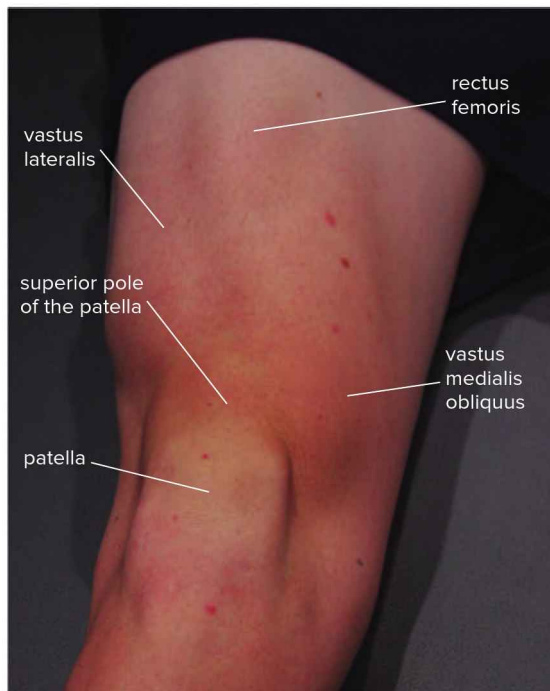
Anterior thigh pain

with ZUZANA MACHOTKA

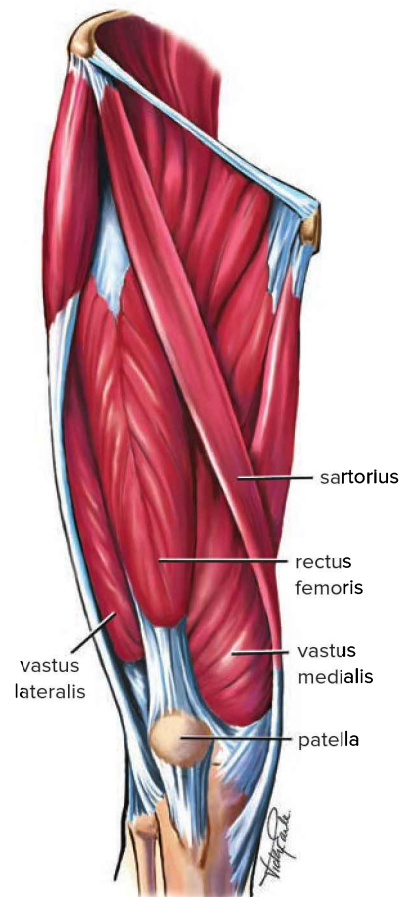
It was the quad in my right leg, the tendon just kept tearing. I couldn't get back to the point of playing and obviously after having a really good start last season, and not playing again, it was frustrating.

Joel Paris, promising Australian cricket fast bowler

The anterior thigh region (Fig. 33.1) is a common site of sporting injuries. Quadriceps muscle strains and contusions constitute the majority of these injuries; however, referred pain from the hip, sacroiliac joint (SIJ) or lumbar spine may also cause anterior thigh pain.¹ Stress fracture of the shaft of the femur is a less common but important diagnosis. A summary of the causes of anterior thigh pain is shown in Table 33.1.

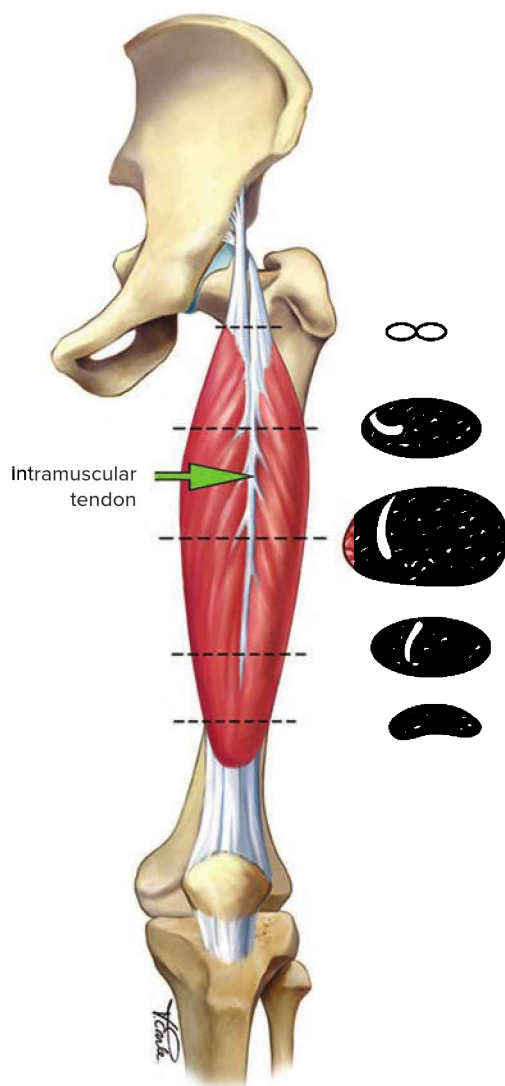


(a)



(b)

Figure 33.1 Anatomy of the anterior thigh (a) Surface anatomy (b) Muscles of the anterior thigh



(c)

Figure 33.1 (cont.) (c) Central tendon of the rectus femoris

EPIDEMIOLOGY

Lower limb injuries, particularly thigh muscle injuries, are common in sport. In athletics, thigh muscle strains account for up to 13.5% of all injuries.¹ In professional soccer, up to a third of injuries involve lower limb muscles and 19% of all muscle injuries involve the quadriceps.^{2, 3} Recurrent quadriceps muscle injuries are associated with a significantly longer period of rehabilitation and time to return to sport. Therefore, an accurate initial diagnosis and a structured rehabilitation program are essential to ensure a safe return to sport.

FUNCTIONAL ANATOMY AND BIOMECHANICS

The thigh musculature is divided into anterior, medial and posterior compartments by intermuscular septa. The anterior compartment is the largest of the three, and includes the femur, quadriceps femoris and sartorius muscles. The quadriceps include the rectus femoris, vastus lateralis, vastus medialis and vastus intermedius muscles (Fig. 33.1b). Concentric contraction of the quadriceps muscles extends the knee against gravity, which is important in such activities as rising from sitting and climbing up stairs. Eccentrically, the quadriceps muscles have an important role in lowering body weight, for example during walking/running down hills and down stairs. The quadriceps muscles contribute enormously to acceleration and power during jumping and running. As a consequence, the quadriceps can be up to three times stronger than their antagonist muscle group, the hamstrings.⁴

The rectus femoris is the only quadriceps muscle to cross the hip joint and has two heads at its origin (Fig. 33.1c). Proximally, the direct head of the rectus femoris originates from the anterior inferior iliac spine and the indirect head originates from above the acetabulum. The two heads form a conjoint tendon just beneath their origin. The direct head is more superficial and has connections with the deep fascia of the thigh (fascia lata). The indirect head contributes to a deep

Table 33.1 Causes of anterior thigh pain

Common	Less common	Not to be missed
Quadriceps muscle contusion ('cork thigh', 'charley horse') Quadriceps muscle strain • distal rectus femoris • proximal rectus femoris Myositis ossificans	Referred pain • upper lumbar spine • sacroiliac joint • hip joint Stress fracture of the femur Sartorius muscle strain Avulsion of the apophysis of rectus femoris Nerve entrapment • lateral cutaneous nerve • femoral cutaneous nerve	Slipped capital femoral epiphysis Perthes' disease (Chapter 44) Tumour (e.g. osteosarcoma of the femur) Acute compartment syndrome of the thigh

musculotendinous junction or 'central tendon', which can travel up to two-thirds of the muscle belly (Fig. 33.1c).⁵⁻⁷ This complex anatomy and its biarthrodial (i.e. two joint) nature, may be why the rectus femoris is the most commonly strained quadriceps muscle.⁸

Distally, all four quadriceps muscles insert into the superior pole of the patella and continue to form the patellar tendon which, in turn, attaches to the tibial tuberosity. Overlying the quadriceps muscles is the sartorius muscle. The sartorius muscle passes medially from the anterior superior iliac spine to the superior/medial surface of the tibia in an arrangement called the pes anserinus (Chapter 37). The sartorius flexes, abducts, and laterally rotates the hip joint, flexes the knee joint and provides restraint to knee valgus forces. Despite its functional importance, the sartorius is less prone to strain injury because it works synergistically with other hip and thigh muscles.

The adductor canal is located deep to the sartorius muscle and extends from the apex of the femoral triangle (Chapter 32) to the adductor hiatus—the gateway to the popliteal fossa. The femoral artery and vein, saphenous nerve, and the nerve to vastus medialis are bundled together within the adductor canal. This site is commonly used to administer anaesthesia (femoral nerve block). Quadriceps weakness can occur following a femoral nerve block and should be monitored postoperatively because the quadriceps can atrophy rapidly following periods of immobilisation or decreased activation.^{9, 10}

CLINICAL APPROACH

In the anterior thigh, primary diagnoses frequently include contusions and strains. However, it is important that clinicians consider important factors that may influence the diagnosis and subsequent rehabilitation. These can include:

- details of previous injuries, including rehabilitation methods, time frames and outcomes
- the magnitude of any impairments in quadriceps strength, control and activation
- any hypermobility or hypomobility, particularly of the hip, knee and ankle joints
- psychological factors such as motivation, pain tolerance and readiness to return to sport.

History

The two most important aspects of the history of a patient with anterior thigh pain are the exact site of the pain and the mechanism of injury. The site of the pain is usually well localised in cases of contusion or muscle strain. Contusions can occur anywhere in the quadriceps muscle but they are most common anterolaterally or in the vastus medialis obliquus. Muscle strains generally occur in the midline of the thigh anteriorly.

The mechanism of injury may help differentiate between muscle strains and contusions. A contusion is likely to be the result of a direct blow, whereas a muscle strain usually occurs when an athlete is striving for extra speed when running or extra distance when kicking. In contact sports, however, the athlete may have difficulty recalling the exact mechanism of injury.

Whether or not the athlete was able to continue activity, the present level of function and range of motion, night pain and the degree of swelling are all guides to the severity of the condition. Determine whether the POLICE regimen was implemented initially (see Chapter 17) and whether there were any aggravating factors (such as continued activity).

Gradual onset of poorly localised anterior thigh pain in a distance runner worsening with activity may indicate a stress fracture of the femur. Burning pain and associated paraesthesia suggest nerve involvement, and poor location of pain can suggest a referred pain from other areas such as the SIJ, hip joint and upper lumbar spine. Bilateral pain suggests the pain is referred from the lumbar spine.

Physical examination

In anterior thigh pain of acute onset, the diagnosis is usually straightforward and examination can focus on local structures. In anterior thigh pain of insidious onset, diagnosis is more difficult and examination should include sites that refer pain to the thigh—the lumbar spine, SIJ and hip.

Muscle tightness and altered muscle activation around the pelvis and thigh can lead to asymmetries further down the kinetic chain. These asymmetries can alter how ground reaction forces are absorbed, particularly during high impact activities such as jumping and sprinting, thus predisposing athletes to further injury. In the patient with anterior thigh pain, assessing and treating muscle imbalances is often necessary to optimise movement patterns and address risk factors for further injury.

The aim of the examination is to determine the exact site of the abnormality and to assess range of motion and muscle strength. Functional testing may be necessary to reproduce the symptoms.

1. Observation (look for muscle bulk, wasting, swelling and position of patella. During activity, observe muscle activation, pelvic control, patella tracking and any signs of a limp)
 - a. standing
 - b. walking
 - c. supine
2. Active movements (are there any asymmetries?)
 - a. hip flexion
 - b. knee flexion
 - c. knee extension
 - d. active straight leg raise¹¹



(a)



(b)

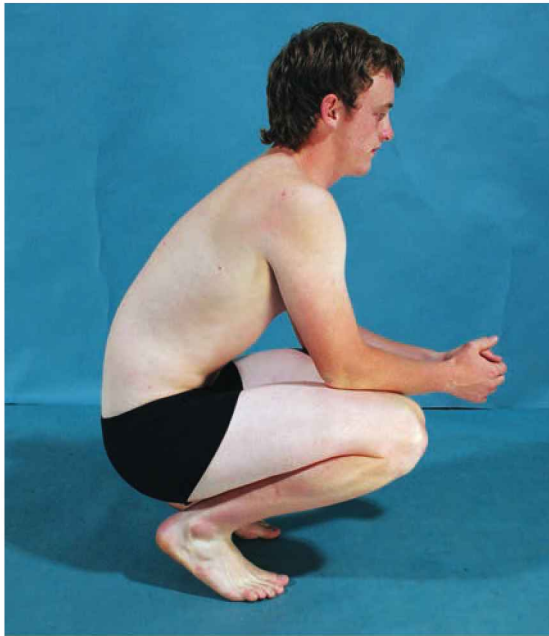


(c)

Figure 33.2 Examination of the patient with anterior thigh pain (a) Passive movement—quadriceps stretch. A passive stretch of the quadriceps muscles is performed to end of range. Passive hip extension may be added to increase the stretch on the rectus femoris, which may reproduce the patient's pain (b) Resisted movement—knee extension. With the hip and knee flexed to 90°, the knee is extended against resistance (c) Resisted movement—hip flexion

3. Passive movements (consider what determines the end feel e.g. muscle spasm)
 - a. hip and knee (e.g. hip quadrant, knee hyper-extension)
 - b. muscle stretch (e.g. quadriceps) (Fig. 33.2a)
 - c. patella glides (Chapter 36)
 - d. Thomas test (Chapter 32)

4. Resisted movements (isometric or concentric. Graded using a 0-5 scale or with a hand-held dynamometer)
 - a. knee extension (Fig. 33.2b)
 - b. straight-leg raise
 - c. hip flexion (Fig. 33.2c)
5. Functional tests
 - a. squat (Fig. 33.2d) (single leg, ¼ squat vs full)
 - b. jump (vertical jumps, box jumps)
 - c. hop (on the spot, hop for distance)^{12, 13}
 - d. kick (short vs long distance)
 - e. lunge (forward vs reverse eccentric lunges)
6. Palpation (consider firmness and if able, map out area of tenderness)
 - a. sartorius and quadriceps muscles (Fig. 33.2e)
 - b. femoral artery (the proximal two-thirds of a line drawn from the mid-inguinal ligament point to the adductor tubercle with the thigh flexed, abducted, and laterally rotated represents the course of the femoral artery and nerve)



(d)



(f)



(e)



(g)

Figure 33.2 (cont.) (d) Functional movements—squat. If the previous activities have failed to reproduce the patient's pain, functional movements should be used to reproduce the pain. These may include squat, hop or jump (e) Palpation—the anterior thigh is palpated for tenderness, swelling and areas of focal muscle thickening. A focal defect in the muscle belly may be palpated, especially with active muscle contraction (f) Special tests—the fulcrum test for the presence of a femoral stress fracture. This is performed with pressure over the distal end of the femur. Reproduction of the patient's pain may be indicative of a femoral stress fracture (g) Special tests—neurodynamic test (modified Thomas test). The patient is placed in the psoas stretch position. Cervical and upper thoracic flexion is added and then the clinician passively bends the patient's knee (using his or her own leg). Reproduction of the patient's symptoms indicates a neural contribution



(h)

Figure 33.2 (cont.) (h) The side-lying slump test—an extension of the standard prone knee bend test and focuses on the neural-sensitive structures of the anterior thigh (e.g. femoral nerve). Test starts with the subject in a side-lying position. Subject holds lower leg for stability in a comfortable, not fully flexed, position with head in cervical flexion. The assessor/therapist stands behind subject supporting the upper leg in a neutral hip position, maintaining neutral hip abduction/adduction throughout the test. With one hand over hip, the other hand is used to support the lower limb in approximately 90 degree knee flexion. Resting symptoms should be recorded (e.g. stretch through anterior thigh). The subject's hip is then extended, noting any changes in resting symptoms and/or position of thigh with onset of symptoms. If no symptoms are evoked, the leg is extended to the point of firm hip resistance. The subject is then asked to extend their neck and report any change in symptoms. The assessor should monitor for any changes in hip resistance before returning to starting position. Test should be repeated on the other side

7. Special tests

- fulcrum test (Fig. 33.2f) (this test is reported to have excellent sensitivity and specificity for mid-shaft femoral stress fractures)^{14, 15}
- neurodynamic testing (modified Thomas test Fig. 33.2g, femoral nerve test,¹⁶ prone knee bend¹⁷, side-lying slump Fig. 33.2h)
- lumbar spine (Chapter 29)
- SIJ (Chapter 30)
- knee jerk reflex (present, strong, endurance)

Key outcome measures

There are currently no patient-reported outcome measures that are specific to anterior thigh pain that can be used to monitor treatment progression. However, the lower

extremity functional scale (LEFS) is a reliable (ICC=0.92) and valid questionnaire that can be used to evaluate function for a variety of musculoskeletal conditions, including anterior thigh pain.^{18, 19} The LEFS involves rating the difficulty of performing 20 functional activities from 0 (extremely difficult) to 4 (no difficulty). The highest possible score of 80 indicates a high level of function. The LEFS has a minimum detectable change of ± 9 points, a standard error of measurement of 4 points and a minimal clinically important difference of ± 9 points.^{18, 19}

It is important to achieve symmetrical quadriceps strength following an anterior thigh injury. The gold standard measure of quadriceps strength is isokinetic dynamometry; however, few clinicians have access to this equipment. Hand-held dynamometers offer a clinically feasible alternative. Isometric quadriceps strength testing with a hand-held dynamometer has been found to be reliable (ICC=0.96) and to correlate moderately with isokinetic testing ($r=0.62$).²⁰ However, due to variability in quadriceps strength between individuals, the minimum detectable change and minimal clinically important difference in quadriceps strength for patients with anterior thigh pain is not known.

Investigations

Investigations are usually not required in athletes with anterior thigh pain, although magnetic resonance imaging (MRI) may have a role in the evaluation of the severity of rectus femoris strains, especially where injuries to the central tendon are suspected.^{5, 6}

If a quadriceps contusion fails to respond to treatment, plain radiographs may demonstrate myositis ossificans. This is usually not evident until at least 3 weeks after the injury. Ultrasound examination or MRI will confirm the presence of a haematoma and may demonstrate early evidence of calcification.

If a stress fracture of the femur is suspected, plain radiograph is indicated. If this is normal, an isotopic bone scan or MRI can be ordered to further rule out the condition.

Although hip pathology most often refers to the groin (Chapter 32), it can refer to the anterior and occasionally lateral, thigh. In adults, hip impingement or osteoarthritis are likely diagnoses; in adolescents, consider a slipped capital femoral epiphysis (Chapter 44) or avulsion fracture, particularly when coupled clinically with an abnormal gait pattern. When thigh pain is associated with restricted or painful hip motion, imaging may be indicated.

QUADRICEPS CONTUSION

Quadriceps contusion, known colloquially as 'charley horse' or 'cork thigh', is an extremely common injury caused by blunt trauma to the thigh. The injury affects both the connective tissue and muscle fibres. As the fibres are disrupted during blunt trauma, the intervening space fills up with blood, which later forms clots.

Table 33.2 Distinguishing features of minor quadriceps contusion and grade I quadriceps muscle strain

Diagnostic features	Quadriceps contusion	Grade I rectus femoris muscle strain
Mechanism	Contact injury	Non-contact
Pain onset	Immediate or soon after	After cool down (next day)
Behaviour of pain (24 hours post-trauma)	Improves with gentle activity	Painful with use
Location	Usually lateral or distal	Rectus femoris muscle belly (proximal or middle-third)
Bruising/swelling	May be obvious early	May be absent or delayed
Palpation findings	Tenderness more obvious, lump may feel ovoid or spherical, becomes progressively harder	May be difficult to find or may be a small area of focal tenderness with a characteristic ring of inflammation surrounding it Muscle spasm in adjacent fibres proximally and distally
Effect of gentle stretch	May initially aggravate pain	Not associated with pain
Strength testing	No loss of strength except pain inhibition	Loss of strength (may need eccentric or functional testing to reproduce pain)

It can sometimes be difficult to distinguish clinically between a contusion and a muscle strain (Table 33.2).

Quadriceps contusion is common in contact sports such as soccer and basketball. In sports such as hockey, lacrosse, and cricket, sporting equipment, such as a hockey stick, ball or puck travelling at high speed may cause a contusion.

Examination confirms an area of tenderness and swelling with worsening pain on active contraction and passive stretch. In severe cases with extensive swelling, pain may be severe enough to interfere with sleep. Assessing the severity of the contusion during the initial assessment is important when planning rehabilitation and return to sport.

Severity can be divided into mild, moderate and severe²¹ and an appropriate time to return to sport can vary from several days to a number of weeks.²¹⁻²³ The degree of passive knee flexion after 24 hours can be used as a clinical indicator of the severity of the haematoma (Table 33.3).²²

In severe cases it is important to identify the exact muscle(s) involved. MRI together with ultrasound are the most appropriate investigations.²⁴ Ultrasonography is useful in determining the precise location of the haematoma.^{25, 26} MRI can provide greater insight into the extent and location of the haematoma and may provide greater prognostic value than ultrasound. An example of an MRI of a severe haematoma is shown in Figure 33.3.

Table 33.3 Grading of quadriceps contusion

Severity	Clinical features	Active knee flexion (°)
Mild	May or may not remember incident Usually can continue activity Sore after cooling down or next morning Normal gait, tender on palpation Minimal loss of strength	>90°
Moderate	Usually remembers incident but can continue activity, although may stiffen up with rest (e.g. half-time or full-time) Some pain on contraction Antalgic gait, tender on palpation May have associated knee effusion	45–90°
Severe	Usually remembers incident May not be able to control rapid onset of swelling/bleeding Difficulty with full weight-bearing Tender over large area (tracking) Obvious bleeding Loss of strength and function	<45°

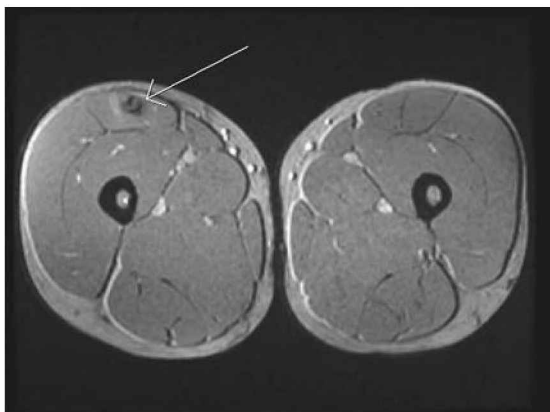


Figure 33.3 MRI appearance of severe haematoma (arrow) of the vastus intermedius muscle

It is important to appreciate that blood from contusions of the lower third of the thigh may track down to the knee joint and irritate the patellofemoral joint. This has been demonstrated in a study of quadriceps contusion in rugby players.²⁷ Quadriceps circumference and the knee brush/swipe test, used to detect swelling around the knee, were associated with greater recovery time, more missed games and a greater number of treatments sought.

Treatment of quadriceps contusion

The most important period in the treatment of a thigh contusion is the first 24 hours following the injury. A player who suffers a thigh contusion should be removed from the field of play and receive the POLICE regimen (Chapter 17). If full weight-bearing is painful, crutches can help unload the muscle and can emphasise the serious nature of the condition to the athlete, team mates and coaches.²² Generally the treatment of a thigh contusion can be divided into four stages:

- stage 1—control of haemorrhage
- stage 2—restoration of pain-free range of motion
- stage 3—functional rehabilitation
- stage 4—graduated return to activity.

A summary of the types of treatment appropriate for each stage is shown in Table 33.4. Progression within each stage, and from one stage to the next, depends on the severity of the contusion and the rate of recovery. For severe contusions, regular ice and compression over 2–3 days should be commenced straightaway. Massage, electrotherapy and stretching should be avoided in the first 3 days. After the initial period, electrotherapy with non-thermal effects (e.g. pulsed ultrasound) may be considered; however, evidence of the effectiveness of such interventions continues to be limited.^{28, 29}



Figure 33.4 Ice and compression treatment of an acute thigh contusion in a position of maximal pain-free stretch

PRACTICE PEARL

In the acute management of a thigh contusion, ice should be applied in a position of maximal pain-free quadriceps stretch (Fig. 33.4). Immobilising the knee in 120° of flexion immediately after injury and for the first 24 hours may be beneficial.³⁰ This is done by bandaging the entire lower limb. The patient then mobilises with crutches.

In the first 7–10 days after a moderate-to-severe contusion there is a considerable risk of re-bleed. The athlete must be careful not to aggravate the bleeding by excessive activity, alcohol ingestion or the application of heat. Care should also be taken with stretching and it should be pain-free in the initial stages. The aim of soft tissue therapy in the first few days after a thigh contusion is to promote lymphatic drainage. Excessively painful soft tissue therapy will cause bleeding to recur and is contraindicated for 48 hours following contusion.

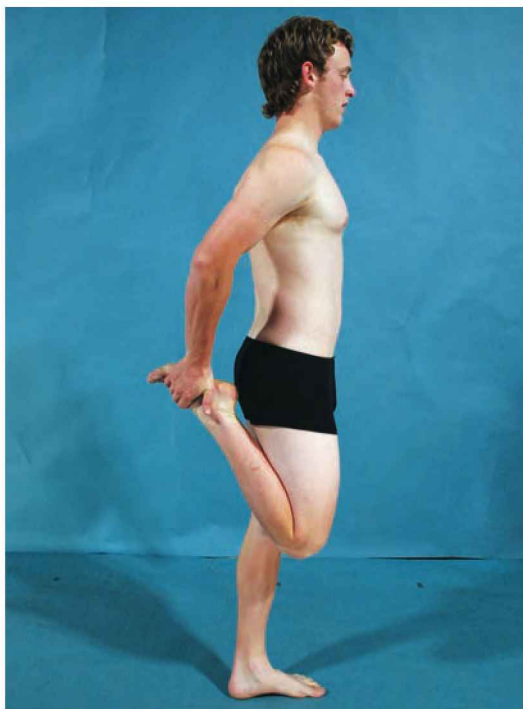
Loss of range of motion is the most significant finding after thigh contusion and range of movement must be regained in a gradual, pain-free progression before return to athletic activity is considered. In a study of military cadets, Ryan et al. found average disability times of 13 days for mild contusions, 19 days for moderate contusions and 21 days for severe contusions.²²

Protective padding, such as thigh pads and athletic pants with removable pads, may help attenuate direct contact forces to the anterior thigh. It has been suggested that athletes in high-risk sports should consider wearing thigh-protective padding routinely.³¹ Players such as ruckmen in Australian rules football, forwards in

Table 33.4 Treatment of moderate quadriceps contusion or grade II muscle strain

Stage	Weight-bearing	Acute management	Soft tissue therapy	Stretching	Strengthening
1. Control haemorrhage	Crutches if FWB	POLICE, compression to include knee joint if lower-third of thigh (Fig. 33.4)	Contraindicated	Gentle active stretch to onset of pain (no hold of stretch) (Fig. 33.5a)	Static muscle contraction if pain-free
2. Restore pain-free range of motion	Progress to WB and FWB as tolerated	Maintain compression bandage when limb is dependent. Ice after exercise	Grade I–II longitudinal gliding away from site of injury Grade II transverse gliding away from site of injury	Increase active stretches to passive stretches to regain ROM	Static muscle contraction, inner-range to through-range (Fig. 33.6a) Stationary exercise: bike Pool: pain-free (walk/swim/kick) Concentric and eccentric exercises (Fig. 33.6b), reverse Nordics (Fig. 33.6c)
3. Functional rehabilitation	FWB	NA	Longitudinal gliding Transverse gliding	Maintain stretch (Fig. 33.5b)	All stage 2 exercises gradually increasing repetitions, speed and resistance Include pulleys, rebounder, proffitter, wall squats, step-downs (Fig. 33.6d) Hopping/jumping, running. Increase eccentric exercises
4. Graduated return to sport	FWB	NA	Myofascial release techniques in knee flexion (Fig. 33.7)		Kicking action with pulleys Multidirectional activities Figure of eight Jumping Plyometrics Graduated specific sporting activities Must complete full training before return to sport Heat-retaining brace may be helpful

ROM = range of motion; PWB = partial weight-bearing; FWB = full weight-bearing



(a)

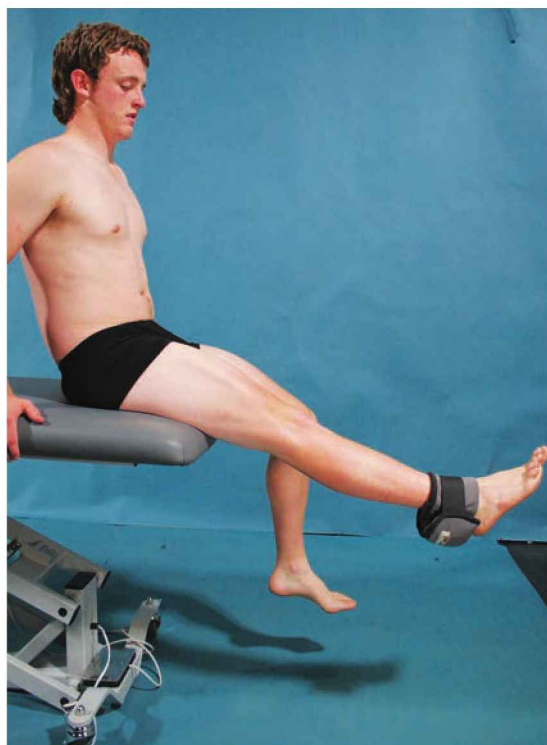


(b)

Figure 33.5 Quadriceps stretching exercises (a) Standard quadriceps stretch while standing. It is important to have good pelvic control and not to lean forward while performing the stretch (b) Passive stretch. The tension of the stretch can be altered by adding hip extension

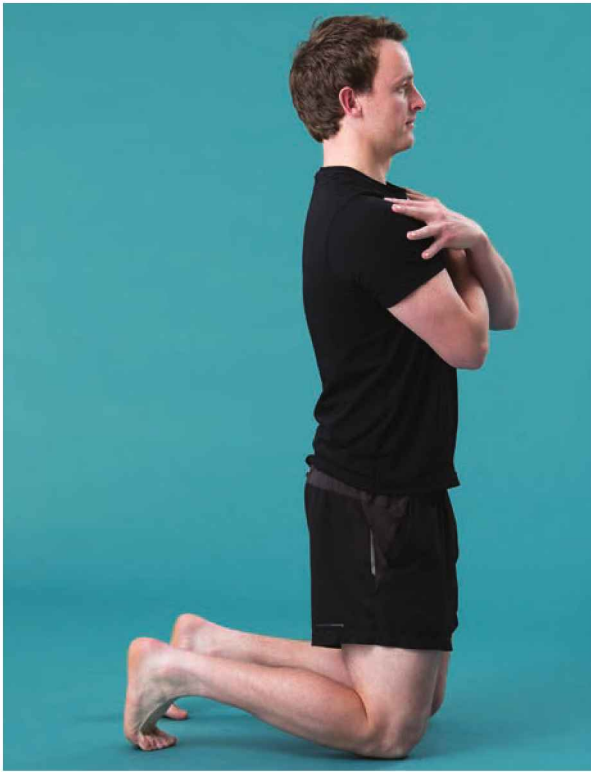


(a)



(b)

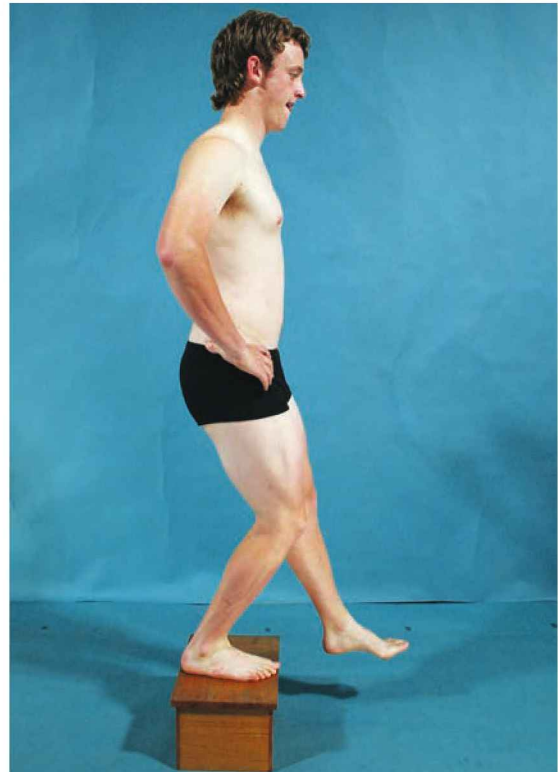
Figure 33.6 Quadriceps strengthening exercises (a) Active quadriceps exercises. Initially inner-range quadriceps strengthening is performed with a rolled towel under the knee as shown. The range is slowly increased, depending on symptoms, until through-range quadriceps contraction can be performed without pain (b) Resisted quadriceps. Concentric and eccentric exercises are performed against gradually increased resistance. Knee extension involves concentric contraction of the quadriceps muscle, while lowering the foot from extension involves eccentric quadriceps contraction



(c-i)



(c-ii)



(d)

Figure 33.6 (cont.) (c) (i) and (ii) Reverse Nordics. Similar to the hamstring Nordic curl exercise but in the opposite direction. Subject/athlete starts in a kneeling upright position (i). The subject/athlete goes as far as they can (ii) until feeling a stretch in anterior thigh or until they cannot maintain neutral trunk position. Focus is placed on the knee being the pivot point while maintaining neutral trunk position throughout exercise (i.e. not breaking at the hips or increasing lumbar lordosis). Exercise can be performed either with ankles in full plantar flexion or on toes. Progressions include arms outstretched with hand weights or arms over head with a medicine ball (d) Functional exercises. A variety of functional exercises can be performed in the late stage of rehabilitation: squats, wall squats, stepdowns (shown), shuttle. Most of these involve both eccentric and concentric contraction of the quadriceps

basketball, and running backs in American football may sustain a series of minor contusions during the course of a game, which may have a cumulative effect and impair performance. Protective padding may help to reduce the severity of contusions; however, evidence for its effectiveness is currently limited.

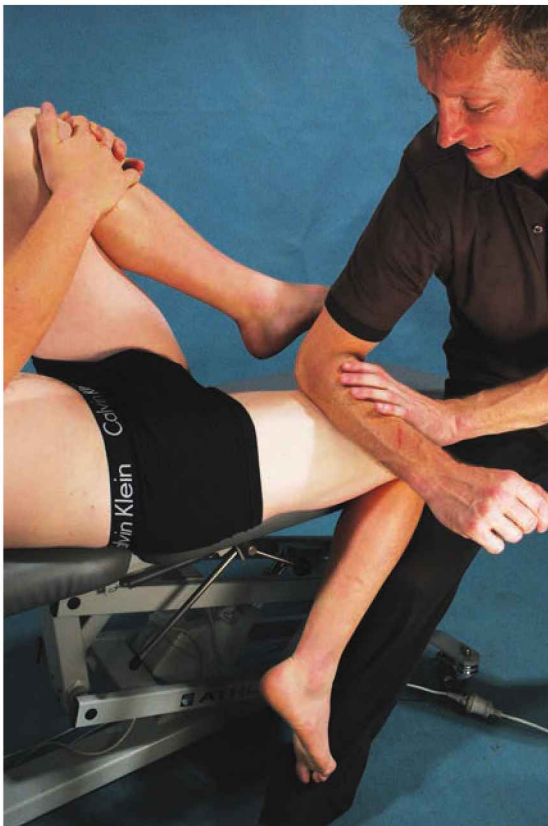


Figure 33.7 Soft tissue therapy—sustained myofascial tension in the position of maximal hip extension and knee flexion

Complications related to contusion

Compartment syndrome of the thigh

Intramuscular haematoma of the thigh after a blunt contusion may result in high intra-compartmental pressures and a diagnosis of compartment syndrome of the thigh. Symptoms often include pain and paraesthesia, and occur with intra-compartmental pressures greater than approximately 20 mmHg.³² Pressures of greater than 30 mmHg over a duration of more than 6 hours can lead to irreversible damage.^{32, 33} Unlike other compartment syndromes, there is no evidence that this condition needs to be treated by surgical decompression unless it occurs in conjunction with other significant injuries or medical conditions, for example fractured femur, coagulation defects.³³ A number of cases of surgery have been reported, but none described necrotic muscle or subsequent clinical evidence of muscle fibrosis, restriction of knee motion or loss of function.³⁴

A case study detailing conservative management of an amateur soccer player who sustained a high-impact injury from an opposing player has been published.³²

Conservative treatment included rest from any lower limb activity for the first 48 hours followed by gentle range of motion exercises of the hip and knee. This routine was slowly progressed over the following months and the athlete returned to soccer after 6 months. Success with conservative treatment can be achievable, but often requires a prolonged rehabilitation period.³²

Myositis ossificans

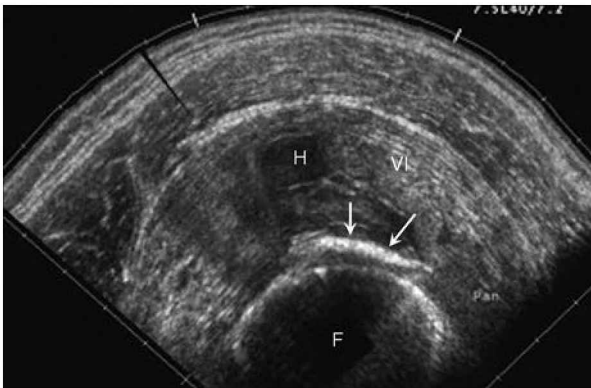
Occasionally, after a thigh contusion, the haematoma calcifies; that is, osteoblasts replace some of the fibroblasts in the healing haematoma and bone tissue is deposited within the muscle. This condition is known as myositis ossificans. The estimated incidence of myositis ossificans following a thigh contusion ranges from 9% to 20%.³⁵ It is unclear why some contusions develop calcification and others do not; however, the more severe the contusion, the more likely that an athlete will develop myositis ossificans.³⁵ Intramuscular contusions appear to be more susceptible to the development of myositis ossificans than intermuscular contusions.

In the acute phase, that is, the first week following the injury, there is a proliferation of fibroblasts into the area of injury.³⁶ The following 10 days, the sub-acute phase, are characterised by the infiltration of osteoblasts. In the final stage, bone is deposited within the muscle. By approximately 6–7 weeks, this bone growth ceases. At this stage a lump is often palpable and a plain radiograph (Chapter 3, Fig. 3.7a) can help to confirm the diagnosis. Ultrasound can demonstrate the myositis ossificans (Fig. 33.8a) however, MRI (Fig. 33.8b) is more sensitive in detecting the inflammatory oedema that precedes myositis ossificans and therefore may be more useful in the early stages.³⁶

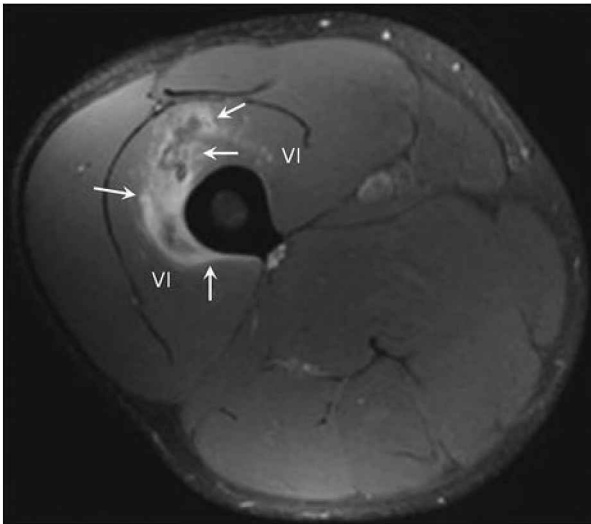
Symptoms of myositis ossificans include ongoing pain in the injured area, morning or night pain, decreased range of motion, local tenderness, stiffness and a palpable mass.³⁷ On palpation, the developing myositis ossificans has a characteristic 'woody' feel. If there is no convincing history of recent trauma, the practitioner must rule out differential diagnoses such as tumours (Chapter 7).³⁸

The risk of myositis ossificans is especially high if the contusion results in prone knee flexion of less than 45° 2–3 days after the injury. Thus, particular care should be taken when managing a severe thigh contusion, as a significant re-bleed may result in the development of myositis ossificans. Inappropriate treatment after muscle injury, such as heat, massage and forced stretching, may increase the risk of developing myositis ossificans.³⁹ The incidence of myositis ossificans appears to be increased when a knee effusion is present.

Once myositis ossificans is established, there is very little that can be done to accelerate the resorption process. Treatment may include local electrotherapy to



(a)



(b)

Figure 33.8 Myositis ossificans (a) Ultrasound appearance 6 weeks after injury showing resolving haemorrhage (H) in the vastus intermedius (VI) with development of ossification (arrows). F = femur (b) MRI appearance with fluid and oedema (arrows) in the vastus intermedius

(A) AND (B) IMAGING IN SPORTS-SPECIFIC MUSCULOSKELETAL INJURIES, 2016, P.81, ALI GUERMAZI, FRANK W ROEMER, MICHEL D CREMA. © SPRINGER INTERNATIONAL PUBLISHING SWITZERLAND 2016. WITH PERMISSION OF SPRINGER

reduce muscle spasm and gentle, painless range of motion exercises. Shock wave therapy has been suggested to improve function for patients with myositis ossificans; however, evidence for its effectiveness is limited.^{40, 41} Acetic acid iontophoresis has been shown to be helpful in two case studies,^{42, 43} while one case study described the intravenous use of bisphosphonate pamidronate.⁴⁴ However, neither method is widely used.

Indomethacin, which reduces new bone formation, has been prescribed as a preventative measure in

high-risk presentations such as severe contusions or those close to bone.³⁵ A corticosteroid injection is absolutely contraindicated in this condition. Surgery is contraindicated in the early stages and only considered when the margins of the ectopic bone are smooth on investigation, suggesting bone maturity. It must be noted that surgery can lead to recurrence post-surgery.³⁵

QUADRICEPS STRAIN

Strains of the quadriceps muscles are common and usually occur during sprinting, jumping or kicking. Fatigue, weakness, muscle imbalance and sprinting/kicking sports are proposed risk factors for quadriceps strains.

In soccer, quadriceps strains are often associated with over-striding when decelerating during running or under-striding during the deceleration phase of the kicking leg when kicking a ball on the run.⁴⁵ Among professional soccer players, quadriceps muscle injuries cause more missed games than hamstring and groin muscle injuries, and re-injury rates are high.³ Strains are most common in the rectus femoris muscle, which is more vulnerable to strain than the other quadriceps muscles, because it passes over two joints—the hip and the knee.

PRACTICE PEARL

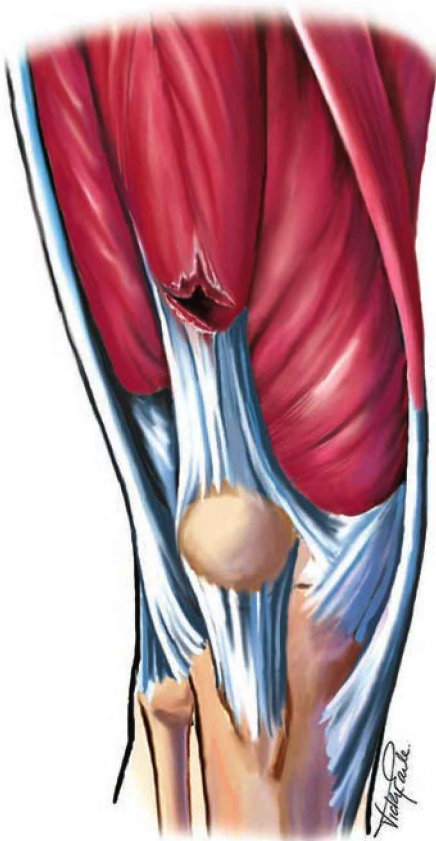
The most common site of strain is the distal musculotendinous junction of the rectus femoris (see below). Management of this type of rectus femoris strain and of strains of the vasti muscles is relatively straightforward; rehabilitation time is short. Strains of the proximal rectus femoris are not as straightforward and are considered in a separate section later in the chapter.

Distal quadriceps muscle strain

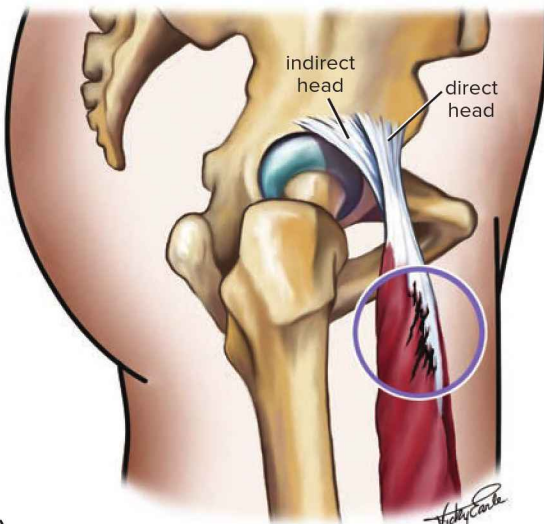
The grading of muscle strains such as distal quadriceps strains (Fig. 33.9a) has recently been reviewed and is discussed in Chapter 3. The athlete feels the injury as a sudden pain in the anterior thigh during an activity requiring explosive muscle contraction. There is local pain and tenderness, pain on stretching and resisted knee extension, and, if the strain is severe, swelling and bruising. If the athlete reports burning or paraesthesia, assessment should rule out femoral nerve injury and neural mechanosensitivity.

Pain on resisted active contraction and on passive stretching are typical signs in quadriceps strains. An area of local spasm is palpable at the site of pain.

Complete tears of the rectus femoris occur with sudden onset of pain and disability during intense



(a)



(b)

Figure 33.9 Two types of quadriceps strains (a) The more common occurs at the distal musculotendinous junction of rectus femoris and has a better prognosis (b) The less common occurs in the proximal rectus femoris and takes longer to repair

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activity. A muscle fibre defect is usually palpable when the muscle is contracted. In the long term, they resolve with conservative management, often with surprisingly little disability.⁴⁶

Treatment

The principles of treatment of a quadriceps muscle strain are similar to those of a thigh contusion. The various treatment techniques shown in Table 33.4 are appropriate for the treatment of quadriceps strain; however, progression through the various stages may be slower for more severe injuries.

Although loss of range of motion may be less obvious for quadriceps strains compared to contusions, it is important that the athlete regains pain-free range of movement as soon as possible. Loss of strength may be more marked than with a thigh contusion and strength retraining should be emphasised in the rehabilitation program, with respect to the stage of healing. As with the general principles of muscle rehabilitation, the program should commence with low-resistance, high-repetition concentric strengthening exercises, and progress to higher resistance and eccentric strengthening exercises.

General fitness can be maintained by activities such as swimming (initially with a pool buoy) and upper body training. Functional retraining such as running and kicking should be incorporated as soon as it is safe to do so. Full training must be completed prior to return to sport. Unfortunately, quadriceps strains often recur; therefore, it is important to maintain an ongoing preventative program.⁴⁷ Prevention should focus on general flexibility of muscles of the thigh and leg, adequate balance of concentric and eccentric strength of the hip flexors and knee extensors and adequate core stability. Exercises based on deceleration under specific sport situations should be included in prevention programs for rectus femoris muscle injuries.

Proximal rectus femoris strain

Strains of the proximal rectus femoris (Fig. 33.9b) can occur either at the conjoint tendon or at the deep musculotendinous junction of the indirect head.^{5, 7, 48} A strain occurring at this musculotendinous junction has a worse prognosis than one occurring peripherally in the muscle belly.

Diagnosis of these deeper injuries is difficult because the physical assessment findings are often inconclusive. MRI can help diagnosis in this instance and demonstrates what is termed a 'bull's eye lesion'⁴⁹ (Fig. 33.10). A bull's eye lesion is present when a disruption is seen around the deep part of the proximal tendon.

Management of proximal rectus femoris strains is influenced by the location and severity of the injury, and the age and athletic demands of the individual. Most proximal rectus femoris strains are managed conservatively and surgical intervention is typically reserved for injuries that involve significant damage to the central tendon (Fig. 33.9). Conservative management aimed at symptom relief and avoidance of re-injury is recommended for most strains.

A study involving 10 professional soccer players reported that players took an average of 3.8 months to return to full training and playing after surgery.⁵⁰ Another study reported that professional Australian rules footballers took an average of 27 days to return to full training following central tendon lesions compared to 9 days for peripheral rectus femoris strains and 4.5 days for strains of the vasti muscles.⁵¹

The use of platelet-rich plasma (PRP) injections has been suggested for these recalcitrant injuries. This treatment is discussed more fully in Chapter 17. There is little evidence for its efficacy in the treatment of muscle injuries.

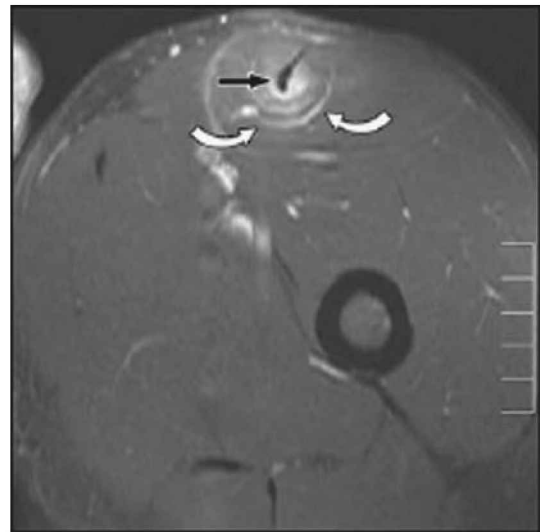
Complete rectus femoris tear

Complete tears of the rectus femoris muscle are uncommon due to the length of the musculotendinous junction (approximately two-thirds of the muscle belly).⁵ In three case studies of complete rectus femoris tears the time from injury to when the athlete sought treatment ranged from 5–7 months.⁵¹

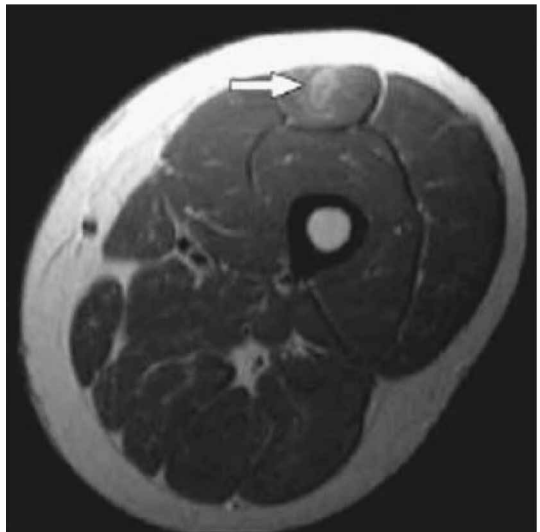
The initial injury may be described as a deep tearing sensation. Patients complain of a tender anterior thigh mass, and weakness and/or pain with activities such as running and kicking. The anterior thigh mass may be associated with muscle retraction.⁵ These signs and symptoms are likely due to the indirect (central tendon) and direct heads of the proximal tendon acting independently, creating a shearing phenomenon in contrast to what occurs in the normal rectus femoris.⁸ Unlike typical strains, which present as focal lesions on MRI, rectus femoris proximal musculotendinous junction injuries have a longitudinal distribution of increased signal along the tendon (Fig. 33.10).⁵

Avulsion injury

Adolescent athletes participating in sports that involve sprinting may sustain an avulsion injury of the anterior inferior iliac spine (AIIS)—the proximal attachment of the direct head of the rectus femoris muscle. These injuries are uncommon in skeletally mature athletes. Patients with greater than 2 cm displacement of the fragment from the AIIS, as determined by X-ray, may require surgery. However, most patients do not require surgery and return to sport after 10–12 weeks.⁵²



(a)



(b)

Figure 33.10 MRI of the proximal rectus femoris musculotendinous junction (a) MRI of a proximal rectus femoris tear in a 19-year-old man shows the characteristic ‘bull’s-eye’ sign; this is made up of a halo of increased signal (white signal highlighted by white arrows) around the deep tendon (black tendon highlighted with a black arrow) (b) A comparable MRI of a similar injury, which differs only in the tendon having a low-signal intensity inside the bright halo (arrow). This is consistent with fatty atrophy in the tendon—the results of the injury having occurred a reasonable time in the past (chronic)

Injury to the indirect head of rectus femoris may be associated with concomitant injury to the hip joint labrum; or a HALTAR lesion (Hip Anterosuperior Labral Tear with Avulsion of Rectus femoris).⁵² This injury has been described in both skeletally mature and skeletally immature athletes and is analogous with the SLAP lesion of the shoulder (Chapter 24). Athletes with rectus femoris injuries who do not progress as expected through rehabilitation may have sustained a concomitant hip labral injury and may require further investigation.

Prevention

Warming up to reduce the risk of injury is common practice in sport. Stretching is commonly performed as part of the warm-up by athletes; however, recent evidence suggests that static stretching does *not* reduce the incidence of exercise-related injury.⁵³⁻⁵⁶

Furthermore, static stretching has been associated with impairments in muscular performance, particularly in the quadriceps and hamstring muscle groups.^{54, 57, 58} On the other hand, studies have shown that dynamic stretching can enhance muscular performance, but as yet, no consensus exists on the most appropriate exercise parameters (e.g. duration and frequency).^{57, 59-61} Dynamic stretches for the anterior thigh muscles can include walking quadriceps stretches, high knee pulls, forward leg swings, walking lunges and various running drills.

In addition to integrating dynamic stretching into the athlete's warm up, stretching and mobility exercises may also need to be addressed after and between training sessions and games. Prevention strategies for rectus femoris injuries include improving general flexibility of muscles of the thigh and leg, and a focus on both concentric and eccentric strength of the hip flexors and knee extensors.

The iliopsoas muscle has a mechanical relationship with the femoral nerve. If the iliopsoas muscle becomes restricted, the mobility of the femoral nerve may also be affected, in the same way that hamstring muscle restrictions can affect the mobility of the sciatic nerve (Chapter 34). Similarly, because the iliopsoas muscle functions synergistically with the rectus femoris in sprinting and kicking, iliopsoas weakness may increase the demand placed on the rectus femoris muscle.⁷

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Flexibility and strengthening programs should address both the rectus femoris and the iliopsoas muscles, rather than considering these muscles in isolation.⁷

In sprinting, the risk for rectus femoris injuries may be highest during the acceleration (eccentric muscle actions in the early swing phase) and deceleration phases. Exercises that simulate this eccentric loading should therefore be included in prevention programs for rectus femoris muscle injuries. Exercises such as the reverse Nordic curl (Fig. 33.6c), forward deceleration and reverse lunge (with/without overhead medicine ball, Fig. 33.11) may improve the capacity of the rectus femoris to tolerate eccentric loads during sport.⁷

Adequate core strength and activation is necessary to counteract torsional forces of the spine, pelvis, and lower limb, particularly during kicking and changing direction. Preventative programs should therefore emphasise strength and activation of the trunk and gluteal muscles, using both isolated and functional tasks.

LESS COMMON CAUSES

Stress fracture of the femur

Stress fractures are common in the athletic population and are categorised as overuse injuries (Chapter 4). Femoral stress fractures can occur around the femoral neck, intertrochanteric and subtrochanteric regions (Chapter 32), and the femoral shaft.⁶² Stress fractures in the distal femur can occur following anterior cruciate ligament (ACL) reconstruction surgery using transfemoral fixation and may be more common in athletes using accelerated rehabilitation programs.^{63, 64} Stress fracture of the shaft of the femur should be suspected in an athlete, especially a long distance runner, who complains of a dull and poorly localised ache in the anterior thigh.¹⁴

Risk factors for developing femoral stress fractures include training errors, hard training surfaces (e.g. concrete), and poor footwear (ill-fitting, worn and inflexible such as military boots).^{62, 65} Intrinsic risk factors include leg length discrepancies and excessive foot pronation or supination.⁶⁵

Pain may be referred to the groin or knee and the athlete may present with an antalgic gait (especially in the case of femoral neck stress fractures).⁶⁶ Clinical assessment should include an assessment of lower limb alignment, particularly leg length and foot posture. Local tenderness, warmth and swelling may be present. The fulcrum test is a pain-provocation test that can be used to confirm the diagnosis (Fig. 33.2f).

Plain radiography may not be sensitive enough to detect changes in bone structure in the initial stages of a femoral stress fracture; however, MRI (Fig. 33.12) has excellent specificity (100%) and sensitivity (86-100%) for identifying femoral stress fractures.⁶⁵ Isotopic bone scans have excellent sensitivity (74-84%) but poor specificity (33%).^{65, 67}



(a)



(b)

Figure 33.11 Reverse lunge with overhead medicine ball: Subject starts in a comfortable split lunge position (a), medicine ball over head. Subject then performs a backwards lunge (b), focusing on placing weight through the behind leg until thigh is vertical. The knee should remain behind the toes of the forward leg

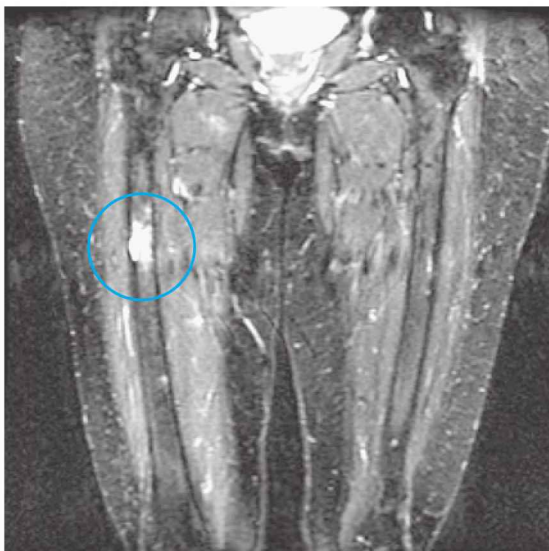


Figure 33.12 MRI of a stress fracture of the shaft of the femur

False positives associated with bone scans include bone infections and tumours. Differentiating between a bone stress reaction and a stress fracture on a bone scan can be challenging because the size of the uptake on the scan may not correlate with the magnitude of the pathology.

Stress fractures are more prevalent among female athletes. The 'female athlete triad', discussed in Chapter 4, is a specific risk factor for stress fractures in females (low energy availability with or without disordered eating, menstrual dysfunction and low bone mineral density).^{68, 69}

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There may be tenderness over the shaft of the femur that can be aggravated if the patient sits with the leg hanging over the edge of a bench, particularly if there is downward pressure placed on the distal femur, the so-called 'hang test' or 'fulcrum test' (Fig. 33.2f).

Treatment of femoral stress fractures depends on the location and type of fracture, but all treatment should initially involve rest from painful activities and maintenance of aerobic fitness with cycling, deep water running or swimming. Weight-bearing as tolerated is often permitted; however, cam boots and crutches may be considered for athletes who are unable to weight bear without pain. Underwater and anti-gravity treadmills may be beneficial for these athletes.⁷⁰ Surgery is indicated where a fracture has become displaced.

When the fulcrum test is completely negative, on average after 7 weeks, it is thought to be safe to initiate return-to-sport progressions in rehabilitation.¹⁵ To avoid associated complications, such as a complete fracture, bone healing must be confirmed before the athlete returns to play.⁹

In females with stress fractures, menstrual disturbance information should be sought and corrected where possible through the aid of bone mineral density tests and endocrine function tests. Nutritional input may be considered for those identified with eating disorders.

Lateral femoral cutaneous nerve injury ('meralgia paraesthetica')

Lateral cutaneous nerve of thigh (LFCN) injury, also known as 'meralgia paraesthetica', can be a cause of anterolateral thigh pain.

Anatomy

The LFCN is a sensory nerve that originates from the lumbar plexus (L2, L3 spinal nerves) and runs along the lateral border of the psoas major muscle, across the iliacus muscle, and exits the abdomen under the inguinal ligament close to the anterior superior iliac spine (ASIS). From here it crosses the sartorius muscle and divides into anterior and posterior branches.

The posterior branch pierces the fascia lata and runs distally supplying the skin of the lateral thigh from the greater trochanter to mid-thigh region. The anterior branch supplies the skin approximately 10 cm below the inguinal ligament and distally to the proximal knee. It should be noted that the anatomical course can have many variations between individuals and is susceptible to compression injury where it exits the pelvis.⁷¹

Clinical biomechanics

Repeated hip flexion and extension during sporting activity (e.g. gymnastics) can irritate the nerve, and entrapment can occur around the ASIS between the ilium and the inguinal ligament.⁷² Obesity and pregnancy increase intra-abdominal pressure and can compress the nerve via the distended abdomen where the nerve leaves the pelvis.⁷³ Tight-fitting garments and athletic

compression garments can lead to symptoms. Entrapment as a result of wearing weight belts in scuba divers has also been reported.⁷⁴

The LFCN is susceptible to injury via blunt trauma around the ASIS and the anterior thigh, especially in contact sports such as rugby.⁷² Repeated falls from a balance beam in gymnastics may also cause injury.⁷⁴ Surgical procedures are a risk factor for LFCN injury, irritation and entrapment.^{32, 75, 76} Injury to the LFCN has been reported following hip arthroscopy because this nerve lies approximately 3 mm from the anterior portal.

Clinical features

Symptoms include pain, burning, numbness, and paraesthesia of the anterolateral thigh without loss of reflex or motor control.^{74, 75} Sporting activity can aggravate symptoms, while sitting may alleviate symptoms by altering the length and tension within the nerve and associated tissues.^{71, 72, 77} Differential diagnoses include femoral nerve injury and referred pain (discussed below). Sensory changes and symptoms exacerbated by hip extension can help to confirm the diagnosis.⁷¹

Management

Treatment is often conservative, with a period of rest and manual therapy until symptoms have resolved. One case series has suggested that Kinesio taping is beneficial, but evidence continues to be limited.⁷⁸ Anti-inflammatory medications can be beneficial in the early phases.⁷⁵ If symptoms do not settle quickly, other interventions such as a local injection of corticosteroid and surgical interventions (either nerve decompression or neurectomy) can be considered, but these interventions currently have limited evidence.⁷³ Thigh and hip pads can be used as preventative strategy in high-contact sports.⁷²

Femoral nerve injury

The femoral nerve passes between the psoas major and the iliacus muscles and exits the abdomen deep to the inguinal ligament through the femoral canal. In the upper thigh, the nerve gives off motor branches to the quadriceps, sartorius and pectineus muscles. The sensory branches supply the skin of the anterior thigh. The femoral nerve then continues distally as the saphenous nerve.

Injury to the nerve can occur secondary to hyperextension of the hip, such as seen in gymnasts, dancers, soccer players, basketball players and long jumpers.^{72, 74} Gymnasts and dancers performing manoeuvres that involve extreme hip extension coupled with knee flexion are susceptible to femoral nerve injury.⁷⁴

Previously, injury was thought to be due to traction placed on the nerve. However, it is now thought to be secondary to muscle strain of the iliopsoas muscle, where

the local haematoma causes compression of the nerve.^{72, 74} Psoas bursitis can also lead to compression and irritation.⁷⁴

Pain is often located around the inguinal region. Reduced power of the knee extensors and/or reduced knee jerk may also be present.^{72, 74} Sometimes, despite motor changes, sensation can be normal.⁷⁴

Conservative treatment is normally trialled first until symptoms settle. Return to sport is possible when strength in the lower limb is regained.

Referred pain

Referred pain may arise from the hip joint, SIJ, lumbar spine (especially upper lumbar) and neural structures. Patients with referred pain may not have a history of injury and may have few signs suggesting local abnormalities. An increase in neural mechanosensitivity may suggest that referred pain is contributing to thigh pain. The modified Thomas' test (Fig. 33.2g) is the most specific neurodynamic tension test for a patient with anterior thigh pain.

If the modified Thomas' test reproduces anterior thigh pain and altering the neural mechanosensitivity (e.g. passive knee flexion/extension) affects the pain, the lumbar spine and psoas muscle should be examined carefully. Any areas of abnormality should be treated and both the local signs (e.g. reproduction of pain with functional testing) and neurodynamic tests should be repeated to assess any changes. The slump knee bend test (side-lying slump) (Fig. 33.2h) has good reliability, excellent sensitivity and specificity, and can be used in patients with suspected lumbosacral radicular pain.^{16, 79} The test is considered positive if the symptoms can be reproduced and if the intensity of the symptoms can be influenced by changing the nerve tension.¹⁶

As with any soft tissue injury, local and referred factors may contribute to the patient's symptoms. Commonly there is hypomobility of the upper lumbar intervertebral



Figure 33.13 Deep soft tissue treatment to the psoas muscle

segments on the affected side associated with a tight psoas muscle. Mobilisation of the hypomobile segments and/or deep soft tissue treatment to the psoas muscle will often reduce symptoms (Fig. 33.13).

REFERENCES

References for this chapter can be found at www.mhhe.com/au/CSM5e

